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### ✓ Div: C3

Q1. A random sample of 200 observations has mean 6.5 cm. Can it be a random sample from a population whose mean is 7 cm and variance is 8.5 cm ?

```
import math
from scipy.stats import norm
pop_m = 7
sam_m = 6.5
v = 8.5
n = 200
std = math.sqrt(v)
z_cal = abs((sam_m - pop_m) / (std / math.sqrt(n)))
z_crit = norm.ppf(0.975)
print("Absolute value of z-calculated is:" , z_cal)
print("Critical z-value is:", z_crit)
if z_cal > z_crit:
    print("Reject the null hypothesis: The sample is unlikely from the given population.")
else:
    print("Fail to reject the null hypothesis: The sample could be from the given population.")
```

```
➦ Absolute value of z-calculated is: 2.4253562503633295
Critical z-value is: 1.959963984540054
Reject the null hypothesis: The sample is unlikely from the given population.
```

Q.2 An examination of the weight of 9 apples provided the following data : 150, 152, 149, 151, 148, 152, 150, 151 and 153 grams. Investigate whether the average weight of the apples can be assumed to be 151 grams.

```
import numpy as np
from scipy.stats import t

data = [150, 152, 149, 151, 148, 152, 150, 151, 153]
n = len(data)
s_mean = np.mean(data)
s_std = np.std(data, ddof=1)
p_mean = 151

t_stat = abs((s_mean - p_mean) / (s_std / np.sqrt(n)))

df = n - 1

t_crit = t.ppf(0.975, df)

print("Sample Mean:", s_mean)
print("Sample Standard Deviation:", s_std)
print("t-Statistic:", t_stat)
print("Critical t-Value:", t_crit)

# Decision
if abs(t_stat) > t_crit:
    print("Reject the null hypothesis: The average weight is significantly different from 151 grams.")
else:
    print("Fail to reject the null hypothesis: The average weight could be 151 grams.")
```

```
➦ Sample Mean: 150.66666666666666
Sample Standard Deviation: 1.5811388300841898
t-Statistic: 0.6324555320336938
Critical t-Value: 2.306004135204166
Fail to reject the null hypothesis: The average weight could be 151 grams.
```

Q.3 Twenty students participated in a mathematics competition. They were provided with additional tutoring sessions for a month before participating in another similar competition. The scores of each student in both competitions were recorded. Test if the scores provided below indicate that the tutoring sessions had a positive impact on the students' performance. Scores in Competition 1: 85, 78, 72, 90, 93, 65, 79, 81, 70, 75, 87, 69, 82, 74, 86, 88, 91, 73, 77, 84 Scores in Competition 2: 88, 80, 75, 91, 95, 68, 82, 84, 73, 79, 89, 71, 85, 77, 90, 92, 94, 76, 78, 83

```

import numpy as np
from scipy.stats import t

sc1 = [85, 78, 72, 90, 93, 65, 79, 81, 70, 75, 87, 69, 82, 74, 86, 88, 91, 73, 77, 84]
sc2 = [88, 80, 75, 91, 95, 68, 82, 84, 73, 79, 89, 71, 85, 77, 90, 92, 94, 76, 78, 83]

dif = np.array(sc2) - np.array(sc1)

m_dif = np.mean(dif)
sd_dif = np.std(dif, ddof=1)
n = len(dif)

t_stat = m_dif / (sd_dif / np.sqrt(n))

df = n - 1

t_crit = t.ppf(0.95, df)

print("Mean of Differences:", m_dif)
print("Standard Deviation of Differences:", sd_dif)
print("t-Statistic:", t_stat)
print("Critical t-Value:", t_crit)

if t_stat > t_crit:
    print("Reject the null hypothesis: Tutoring had a positive impact.")
else:
    print("Fail to reject the null hypothesis: No significant evidence of a positive impact.")

```

 Mean of Differences: 2.55  
 Standard Deviation of Differences: 1.190974832912761  
 t-Statistic: 9.575304506946091  
 Critical t-Value: 1.729132811521367  
 Reject the null hypothesis: Tutoring had a positive impact.

Q.4 Test the significance of the difference between the means of two normal population with the same standard deviation from the following data. Size Mean St. Dev Sample-1 1000 25 5 Sample-2 2000 23 7

```

import math
from scipy.stats import norm

m1 = 25
m2 = 23
s1 = 5
s2 = 7
n1 = 1000
n2 = 2000

sp = math.sqrt(((n1 - 1) * s1**2 + (n2 - 1) * s2**2) / (n1 + n2 - 2))

z_stat = (m1 - m2) / math.sqrt((sp**2 / n1) + (sp**2 / n2))

z_crit = norm.ppf(0.975)

print("Pooled Standard Deviation:", sp)
print("Z-Statistic:", z_stat)
print("Critical Z-Value:", z_crit)

if abs(z_stat) > z_crit:
    print("Reject the null hypothesis: Significant difference between the means.")
else:
    print("Fail to reject the null hypothesis: No significant difference between the means.")

```

 Pooled Standard Deviation: 6.403332604638808  
 Z-Statistic: 8.064515954086545  
 Critical Z-Value: 1.959963984540054  
 Reject the null hypothesis: Significant difference between the means.

Q.5. The weights of eight randomly selected athletes are recorded in kilograms: 70, 75, 78, 80, 82, 85, 87, 90. The weights of twelve randomly selected basketball players are recorded in kilograms: 72, 74, 76, 78, 79, 80, 82, 83, 84, 85, 87, 88. Can it be concluded that basketball players, on average, weigh more than athletes?

```

import numpy as np
from scipy.stats import t

a_wt = [70, 75, 78, 80, 82, 85, 87, 90]
b_wt = [72, 74, 76, 78, 79, 80, 82, 83, 84, 85, 87, 88]

m_a = np.mean(a_wt)

```

```

m_o = np.mean(o_wt)
sd_a = np.std(a_wt, ddof=1)
sd_b = np.std(b_wt, ddof=1)
n1 = len(a_wt)
n2 = len(b_wt)

t_stat = (m_b - m_a) / np.sqrt((sd_a**2 / n1) + (sd_b**2 / n2))

df = ((sd_a**2 / n1 + sd_b**2 / n2)**2) / \
      ((sd_a**2 / n1)**2 / (n1 - 1) + (sd_b**2 / n2)**2 / (n2 - 1))

t_crit = t.ppf(0.95, df)

print("Mean Weight (Athletes):", m_a)
print("Mean Weight (Basketball Players):", m_b)
print("t-Statistic:", t_stat)
print("Critical t-Value:", t_crit)

if t_stat > t_crit:
    print("Reject the null hypothesis: Basketball players weigh significantly more.")
else:
    print("Fail to reject the null hypothesis: No significant evidence that basketball players weigh more.")

↩ Mean Weight (Athletes): 80.875
Mean Weight (Basketball Players): 80.66666666666667
t-Statistic: -0.07600075519463105
Critical t-Value: 1.7771081745259885
Fail to reject the null hypothesis: No significant evidence that basketball players weigh more.

```